STACKS, DERIVED CATEGORIES OF SPACES: EXACT-SEQUENCE-J-STAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1888-1910 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 1912-1976, the complete author proof (63 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_spaceperfect.json (pb=2350, stmt 1888-1910, proof 1912-1976).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-exact-sequence-j-star' also exists in stacks-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0152: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1976–1976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 014Z: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1976–1976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 079P: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1976–1976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GV: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1823–1841. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1541–1555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01FW: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1976–1976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1976–1976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(U \subset X, V \to X)$ be an elementary distinguished square of algebraic spaces over S.*

(1) *For every sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ we have a short exact sequence*
$$0 \to \mathcal{F} \to j_{U,*}\mathcal{F}|_U \oplus j_{V,*}\mathcal{F}|_V \to j_{U\times_X V,*}\mathcal{F}|_{U\times_X V} \to 0$$
(2) *For any object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*
$$E \to Rj_{U,*}E|_U \oplus Rj_{V,*}E|_V \to Rj_{U\times_X V,*}E|_{U\times_X V} \to E[1]$$
in $D(\mathcal{O}_X)$.

Proof. Let W be an object of $X_{étale}$. We claim the sequence
$$0 \to \mathcal{F}(W) \to \mathcal{F}(W \times_X U) \oplus \mathcal{F}(W \times_X V) \to \mathcal{F}(W \times_X U \times_X V)$$
is exact and that an element of the last group can locally on W be lifted to the middle one. By Lemma 08GN the pair $(W \times_X U \subset W, V \times_X W \to W)$ is an elementary distinguished square. Thus we may assume $W = X$ and it suffices to prove the same thing for
$$0 \to \mathcal{F}(X) \to \mathcal{F}(U) \oplus \mathcal{F}(V) \to \mathcal{F}(U \times_X V)$$
We have seen that
$$0 \to j_{U\times_X V!}\mathcal{O}_{U\times_X V} \to j_{U!}\mathcal{O}_U \oplus j_{V!}\mathcal{O}_V \to \mathcal{O}_X \to 0$$
is a exact sequence of $\mathcal{O}_X$-modules in Lemma 08GV and applying the right exact functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathcal{F})$ gives the sequence above. This also means that the obstruction to lifting $s \in \mathcal{F}(U \times_X V)$ to an element of $\mathcal{F}(U) \oplus \mathcal{F}(V)$ lies in $\mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) = H^1(X, \mathcal{F})$. By locality of cohomology (Cohomology on Sites, Lemma 01FW) this obstruction vanishes étale locally on X and the proof of (1) is complete.

Proof of (2). Choose a K-injective complex $\mathcal{I}^\bullet$ representing E whose terms $\mathcal{I}^n$ are injective objects of $Mod(\mathcal{O}_X)$, see Injectives, Theorem 079P. Then $\mathcal{I}^\bullet|U$ is a K-injective complex (Cohomology on Sites, Lemma 08FI). Hence $Rj_{U,*}E|_U$ is represented by $j_{U,*}\mathcal{I}^\bullet|_U$. Similarly for V and $U \times_X V$. Hence the distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes
$$0 \to \mathcal{I}^\bullet \to j_{U,*}\mathcal{I}^\bullet|_U \oplus j_{V,*}\mathcal{I}^\bullet|_V \to j_{U\times_X V,*}\mathcal{I}^\bullet|_{U\times_X V} \to 0.$$
This sequence is exact by (1). □